

Marwen Ben Khemis

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Florianopolis, Brazil - relocating by August 31, 2026

PROFESSIONAL SUMMARY

Product and software engineer with 5+ years building operator-facing software across backend, web, mobile, and accountable AI workflows. Current Forward Deployed Engineer with Python, PostgreSQL, TypeScript, React admin tooling, and workflow automation experience, including Karta's production customs platform and post-declaration AI system. Built Jerbien, a retail platform that handled 10M+ transactions, and shipped product decisions from field discovery through rollout, monitoring, and production support.

CORE COMPETENCIES

Product Engineering · Python & PostgreSQL Services · React Admin Tooling · Workflow Automation & AI Gates · Customer Discovery & Delivery · Operational Reliability · REST/OpenAPI Integrations · Founder-Level Ownership

WORK EXPERIENCE

Karta / DJAM DKS Jul 2025 - Present

Forward Deployed Engineer

Remote, then on-site

- Built Karta from concept to production in about four months; it has processed 4,000+ customs transactions worth more than \$20M and saved 60+ work-days.
- Owned workflow discovery, Python/PostgreSQL services, typed validation, background workers, React admin tooling, document generation, dashboards, and production support for customs operators.
- Designed AI-assisted workflows with deterministic tools, human approval gates, audit logs, replay, and shadow-mode evaluation; a flagship specialist reached 89.7% step accuracy and 83.9% end-to-end completion without operator override.
- Added live queues, case monitoring, worker health checks, append-only audit trails, and recovery controls for teams working under fixed operational deadlines.

Makhabzti Digitale Oct 2020 - Jan 2026

Founder & CEO

Tunisia

- Founded a retail-technology company that reached 15+ bakeries in under three months, supported 15,000+ TND in daily business volume, and earned Tunisia's national Startup Label.
- Built Jerbien into a production POS and operations platform that handled 10M+ transactions and cut cashier checkout time by about 80% by redesigning workflows around real counter behavior.
- Delivered a TypeScript/Node.js and PostgreSQL backend, Express REST/OpenAPI services, Nuxt/Vue merchant portal, React admin tools, Flutter POS clients, and generated Dart/TypeScript SDKs.
- Protected sales, refunds, offline replay, and multi-tender payments with atomic writes, idempotency, advisory locks, exact decimal money handling, ledger-safe reversals, and typed validation.
- Shipped FOURN'OR for invoicing, tax rules, discounts, catalogues, receipts, seller accounts, and desktop installation.
- Ran production delivery with Docker, Kubernetes, GitHub Actions, ingress/TLS, health probes, Sentry, and Prometheus, and mentored 50+ interns through completed placements.

Moojup May 2025 - Jul 2025

Flutter SDK Engineer Intern / IT Specialist (Payments)

Tunisia

- Built a reusable Flutter SDK for Verifone X990 payment terminals, exposing Android AIDL hardware capabilities through a typed Dart API and Flutter method channels.
- Defined reusable integration boundaries so partner teams could connect terminal functions without rebuilding the native bridge.

Data Engineering Intern / Data Analyst

Manouba, Tunisia

- Built a PostgreSQL warehouse and Python ETL pipelines from MongoDB, then delivered Power BI dashboards for analysis.
- Deployed the data stack through Docker, Kubernetes, ArgoCD, and CI/CD workflows.

EDUCATION

Master's in Business Analytics South Mediterranean University (MSB) 2024 - 2026

Graduated with Outstanding Distinction. Machine learning, econometrics, forecasting, data management, Python analytics, ERP/SAP, and business strategy.

Bachelor's in Data Science ISAMM, Manouba 2019 - 2023

Big data, machine learning, Python, R, SQL/NoSQL, data visualization, and Agile/Scrum.

RECOGNITION & PROGRAMS

Tunisia National Startup Label	Makhabzti Digitale	
Entrepreneur Program, Season 4	Westerwelle Startup House	2022
PMI-PBA / IIBA-PBA Certification Training	Tenstep Tunisia	2023
Volunteer Trainer, Mobile Development & Technology Entrepreneurship	University Technical Clubs	2020 - 2021

SKILLS

Backend & Data: Python, TypeScript, Node.js, PostgreSQL, MongoDB, Redis, REST/OpenAPI, typed validation, background workers, ETL
Product & Frontend: React admin tooling, Nuxt/Vue, Flutter, product discovery, operator workflows, document generation, dashboards
AI & Reliability: LLM-backed workflows, deterministic tools, human approval gates, replay, shadow-mode evaluation, audit trails, monitoring
Delivery & Infrastructure: Docker, Kubernetes, GitHub Actions, Linux, Sentry, Prometheus, production support
Languages: Arabic (native), English (C1 professional), French (full professional), German (A2), Portuguese (beginner, actively learning)